

Higher social tolerance is associated with more complex facial behavior in macaques

Reviewed Preprint

Revised by authors after peer review.

About eLife's process

Reviewed preprint version 2

August 31, 2023 (this version)

Reviewed preprint version 1

May 5, 2023

Sent for peer review

February 23, 2023

Posted to bioRxiv

February 14, 2023

Alan V. Rincon , Bridget M. Waller, Julie Duboscq, Alexander Mielke, Claire Pérez, Peter R. Clark, Jérôme Micheletta 

Department of Psychology, Centre for Comparative and Evolutionary Psychology, University of Portsmouth, Portsmouth PO1 2DY, UK • Centre for Interdisciplinary Research on Social Interaction, Department of Psychology, Nottingham Trent University, Nottingham NG1 4BU, UK • UMR7206 Eco-Anthropology, CNRS-MNHN-Université Paris Cité, Paris 75016, France • Origins of Mind Group, School of Psychology and Neuroscience, University of St Andrews, St Andrews KY16 9JP, UK • School of Psychology, University of Lincoln, Lincoln LN6 7TS, UK

 https://en.wikipedia.org/wiki/Open_access

 <https://creativecommons.org/licenses/by/4.0/>

Abstract

The social complexity hypothesis for communicative complexity posits that animal societies with more complex social systems require more complex communication systems. We tested the social complexity hypothesis on three macaque species that vary in their degree of social tolerance and complexity. We coded facial behavior in >3000 social interactions across three social contexts (aggressive, submissive, affiliative) in 389 animals, using the Facial Action Coding System for macaques (MaqFACS). We quantified communicative complexity using three measures of uncertainty: entropy, specificity, and prediction error. We found that the relative entropy of facial behavior was higher for the more tolerant crested macaques as compared to the less tolerant Barbary and rhesus macaques across all social contexts, indicating that crested macaques more frequently use a higher diversity of facial behavior. The context specificity of facial behavior was higher in rhesus as compared to Barbary and crested macaques, demonstrating that Barbary and crested macaques used facial behavior more flexibly across different social contexts. Finally, a random forest classifier predicted social context from facial behavior with highest accuracy for rhesus and lowest for crested, indicating there is higher uncertainty and complexity in the facial behavior of crested macaques. Overall, our results support the social complexity hypothesis.

eLife assessment

This study shows **important** evidence of the correlation between social tolerance and communicative complexity in a comparison of three macaque species. Notably, the authors use an innovative, detailed methodology for quantifying facial expressions during social interactions. The results are **convincing** regarding a positive association between social complexity and facial behaviour, which should stimulate further comparative research in this field.

Introduction

Animals must overcome a range of environmental and ecological challenges to survive and reproduce, with group-living species having to overcome additional social challenges to maximize fitness. Communicative signals can be used to navigate a number of different social situations and may need to become more elaborate as social complexity increases. The social complexity hypothesis for communicative complexity encapsulates this idea, proposing that animal societies with more complex social systems require more complex communication systems [1].

The social complexity hypothesis has become a topical issue in recent years, with questions regarding the definitions, measurement, and selective pressures driving both social and communicative complexity [2,3]. Social complexity as experienced by group members can be affected by the level of differentiation of social relationships, where complexity increases as social relationships become more differentiated [4,5]. In a socially complex society individuals interact frequently with each other in diverse ways and in many different contexts [1]. If the types of interactions that individuals have is constrained, for example, by dominance or kinship, then social complexity decreases [1]. Social complexity is also affected by the predictability or consistency of social interactions [5,6]. When the behavior of social partners is unpredictable, such as when the dominance hierarchy is unstable, individuals likely perceive the social environment as more complex [6]. These operational definitions of social complexity are valuable to advance the study of social complexity but are not easy to quantify with a single measure [7].

Similarly, communicative complexity is also difficult to quantify. Many studies have used the number of signalling units as a measure of communicative complexity [2]. While a useful measure, it is not always apparent what a signaling unit is. For example, calls are sometimes graded on a continuous scale without a clear separation between different call types [8]. Fewer studies have investigated the complexity of non-vocal communication [1,2], but similar issues exist. One previous study quantified the repertoire of facial behavior in macaques by the number of discrete facial expressions that a species displays and found that it was positively correlated with conciliatory tendency and counter-aggression across species [9]. However, classifying facial expressions into discrete categories (e.g., bared-teeth display) does not capture the full range of expressiveness and meanings that the face can convey. For example, subtle morphological variations in bared-teeth displays are associated with different outcomes of social interactions (e.g., affiliation versus submission) in crested macaques (*Macaca nigra*) [10]. A better approach is to quantify facial behavior at the level of individual facial muscle movements [11], which can be done using the Facial Action Coding System (FACS) [12]. In FACS, visible muscle contractions in the face are called Action Units and allow for a detailed and objective description of facial behavior [11,12]. Indeed, facial mobility, as defined by the number of Action Units that a species has, is positively correlated with group size across non-human primates [13]. However, isolated muscle movements still do not account for the full diversity of facial behavior because facial muscles often contract simultaneously to produce a large variety of distinct facial expressions.

One promising avenue to approximate complexity in living organisms is to quantify the uncertainty or predictability of a system [14,15], which are general properties of complex systems [16,17]. Shannon's information entropy [18] is a measure of uncertainty that can be applied to animal communication. Conceptually, entropy measures the potential amount of information that a communication system holds, rather than what is actually communicated [18,19]. Entropy increases along two dimensions: (i) with increasing diversity of signals, and (ii) as the relative frequency of signal use becomes more balanced. For example, a system with three calls can hold more information than a system with one call and thus would have higher

entropy. Likewise, a system with three calls used with equal frequency will have a higher entropy than another system that expresses one call more frequently than the two others. Uncertainty increases with entropy because each communicative event has the potential to derive from a greater number of units. The relative entropy, or uncertainty, of different systems can be compared by calculating the ratio between the observed and maximum entropy of each system.

The predictability and uncertainty of a communication system is also affected by how flexibly signals are used across different social contexts [5]. For instance, if signal A is always used in an aggressive context and signal B is always used in an affiliative context, then it is easy to predict the context from the signal. Conversely, if signals A and B are used in both contexts, then predictability is lower, and complexity is higher. Extremely rare signals do not substantially affect the predictability of a system regardless of whether they have high or low specificity since they are seldom observed in the majority of social interactions. Therefore, predictability is highest when signals are both highly context-specific and occur in that context often. Additionally, predictability can be measured directly by training a machine learning classifier to predict the social context that a given signal was used in. Differences in prediction error would approximate the relative uncertainty and complexity, with accuracy being lower in more complex systems. However, as complexity lies somewhere between order and randomness [15,19], we should still be able to predict the social contexts better than chance, even in a complex system.

Studying closely related species offers a robust means of testing the social complexity hypothesis due to their homologous communication systems. For this reason, macaques (genus *Macaca*) are excellent taxa to test the social complexity hypothesis. All species have a similar social organization consisting of multi-male, multi-female groups, but vary in social style in ways that are highly relevant to predictions of the social complexity hypothesis. The social styles of macaques consist of several covarying traits that can be ordered along a social tolerance scale ranging from the least (grade 1) to most tolerant (grade 4) [20,21]. Social interactions for the least tolerant species, such as rhesus (*M. mulatta*) and Japanese (*M. fuscata*) macaques, are generally more constrained by a steep linear dominance hierarchy [22] and nepotism [23–25]. Additionally, severe agonistic interactions are more frequent [25], instances of counter-aggression and reconciliation after conflicts are rare [22,25], and formal signals of submission are commonly used [26,27]. Combined, these behavioral traits indicate that agonistic interactions of the least tolerant species are more stereo-typed and formalized. Thus, the outcome of such interactions is more certain, whereas the opposite is true for the most tolerant species, such as crested and Tonkean (*M. tonkeana*) macaques. The unpredictability in the outcome of agonistic interactions of tolerant macaques potentially results in a social environment that is perceived as more complex by individuals [6], where more subtle means of negotiation during conflicts may be necessary.

In this study we compared the facial behavior of three macaque species that vary in their degree of social tolerance and, therefore, social complexity: rhesus (least tolerant), Barbary (*M. sylvanus*, mid-tolerant), and crested macaques (most tolerant). For macaques (and primates in general), the face is central to communication and is a key tool in allowing individuals to achieve their social goals by communicating motivations, emotions and/or intentions [28,29]. We coded facial behavior at the level of individual visible muscle movements using FACS and recorded all observed unique combinations, rather than classifying facial expressions into discrete categories. Based on the social complexity hypothesis [1], we expected that tolerant species would have higher communicative complexity, given that their social relationships are less constrained by dominance and have higher overall uncertainty in the outcome of agonistic interactions. Specifically, we predicted the following: (1) relative entropy of facial behavior will be lowest in the rhesus and highest in crested macaques, (2) context specificity of facial behavior will be highest in rhesus and lowest in crested macaques, and (3) social context can be predicted from facial behavior most accurately in rhesus and least accurately in crested macaques. For all three metrics, we expected Barbary macaques to lie somewhere in between the rhesus and crested macaques.

Entropy of facial behavior

To compare the relative uncertainty in the facial behavior of macaques, we defined facial behavior by the unique combination of Action Units (facial muscle movements) that occurred at the same time. We calculated the entropy ratio for each species and social context, defined as the ratio between the observed entropy and the expected entropy if Action Units were used randomly. Values closer to 0 indicate that there is low uncertainty (e.g., when only a few facial movements are used frequently) and values closer to 1 indicate high uncertainty (e.g., when many facial movements are used frequently). To determine whether the entropy ratios for each species differed within social context, we calculated the entropy ratio on 100 bootstrapped samples of the data, resulting in a distribution of possible values. The bootstrapped entropy ratio of facial behavior differed across species and within social contexts (**Figure 1**). In an affiliative context, the entropy ratio was highest for crested, then Barbary, and lowest for rhesus macaques (crested: mean = 0.52, range = 0.50–0.53; Barbary: mean = 0.45, range = 0.45–0.46; rhesus: mean = 0.38, range = 0.37–0.39). In an aggressive context, the entropy ratio was highest for crested, then rhesus and lowest for Barbary macaques (crested: mean = 0.62, range = 0.60–0.65; Barbary: mean = 0.32, range = 0.32–0.33; rhesus: mean = 0.48, range = 0.47–0.49). In a submissive context, the entropy ratio was highest for crested, then Barbary, and lowest for rhesus macaques. (crested: mean = 0.67, range = 0.64–0.70; Barbary: mean = 0.49, range = 0.48–0.50; rhesus: mean = 0.38, range = 0.37–0.39). Overall, across all contexts, including when the context was unclear, the entropy ratio was highest for crested, and similar for Barbary and rhesus macaques (crested: mean = 0.57, range = 0.56–0.58; Barbary: mean = 0.51, range = 0.51–0.51; rhesus: mean = 0.52, range = 0.51–0.52; **Figure 1**).

Context specificity of facial behavior

We calculated the context specificity for all possible combinations of Action Units. Here we report specificity for combinations that were observed in at least 1% of observations per species and social context because extremely rare signals do not affect the predictability of a system substantially, regardless of whether they have high or low specificity. Specificity for each Action Unit combination was defined as the number of times it was observed in one context divided by the total number of times it was observed across all contexts. When considering single Action Units, some were observed in only one context, but most were observed at least once in all three contexts for all three species (**Figure 2**). On average, single Action Units were observed in fewer contexts for rhesus (mean degree = 1.9), compared to Barbary (mean degree = 2.4), and crested macaques (mean degree = 2.6). The specificity of all Action Unit combinations used in an affiliative context was highest for the rhesus macaques, then Barbary, and lowest for crested macaques (rhesus: mean = 0.80, SD = 0.28, n = 69; Barbary: mean = 0.63, SD = 0.26, n = 450; crested: mean = 0.37, SD = 0.26, n = 327; **Figure 3a**). The specificity of Action Unit combinations used in an aggressive context was highest for rhesus, then crested, and lowest for Barbary macaques (rhesus: mean = 0.71, SD = 0.35, n = 83; Barbary: mean = 0.44, SD = 0.38, n = 64; crested: mean = 0.51, SD = 0.30, n = 281). The specificity of Action Unit combinations used in a submissive context was also highest for rhesus, then crested, and lowest for Barbary macaques (rhesus: mean = 0.93, SD = 0.18, n = 312; Barbary: mean = 0.61, SD = 0.18, n = 297; crested: mean = 0.70, SD = 0.21, n = 595). The majority (>50%) of Action Unit combinations used by rhesus macaques had high specificity (>0.8) in all three social contexts, whereas only a minority (<50%) of Action Unit combinations used by Barbary and crested macaques had high specificity (**Figure 3b**).

Predicting social context from facial behavior

A random forest classifier was able to predict social context (affiliative, aggressive or submissive) from facial behavior with a better accuracy than expected by chance alone for all three species of macaques. The classifier was most accurate for rhesus (kappa = 0.92), then Barbary (kappa = 0.68),

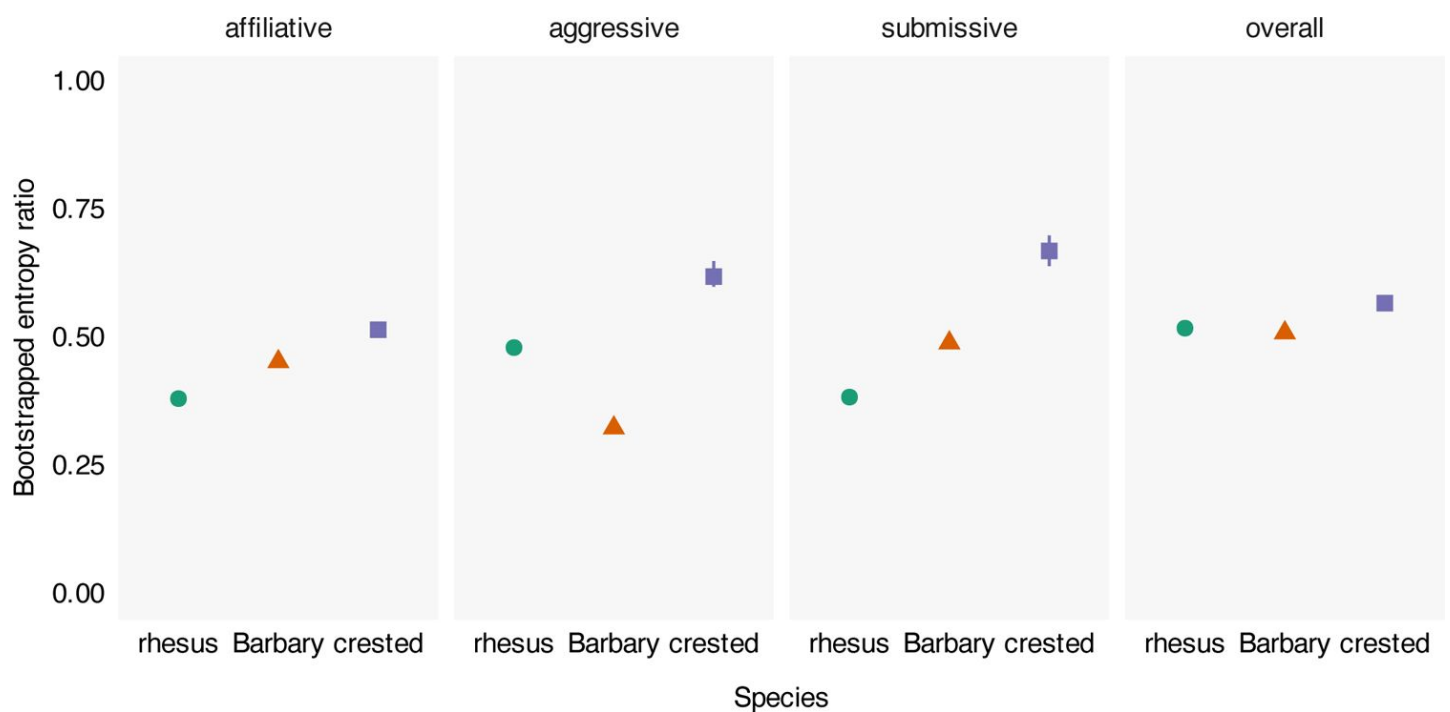


Figure 1

Bootstrapped entropy ratio of facial behavior across social contexts for three species of macaques. The entropy ratio was calculated on 100 bootstrapped samples of the data by dividing the observed entropy by the expected entropy if Action Units were used randomly for each social context. The entropy ratio ranges from 0 to 1, with higher values indicating higher uncertainty. Symbols and whiskers indicate mean and range of bootstrapped values.

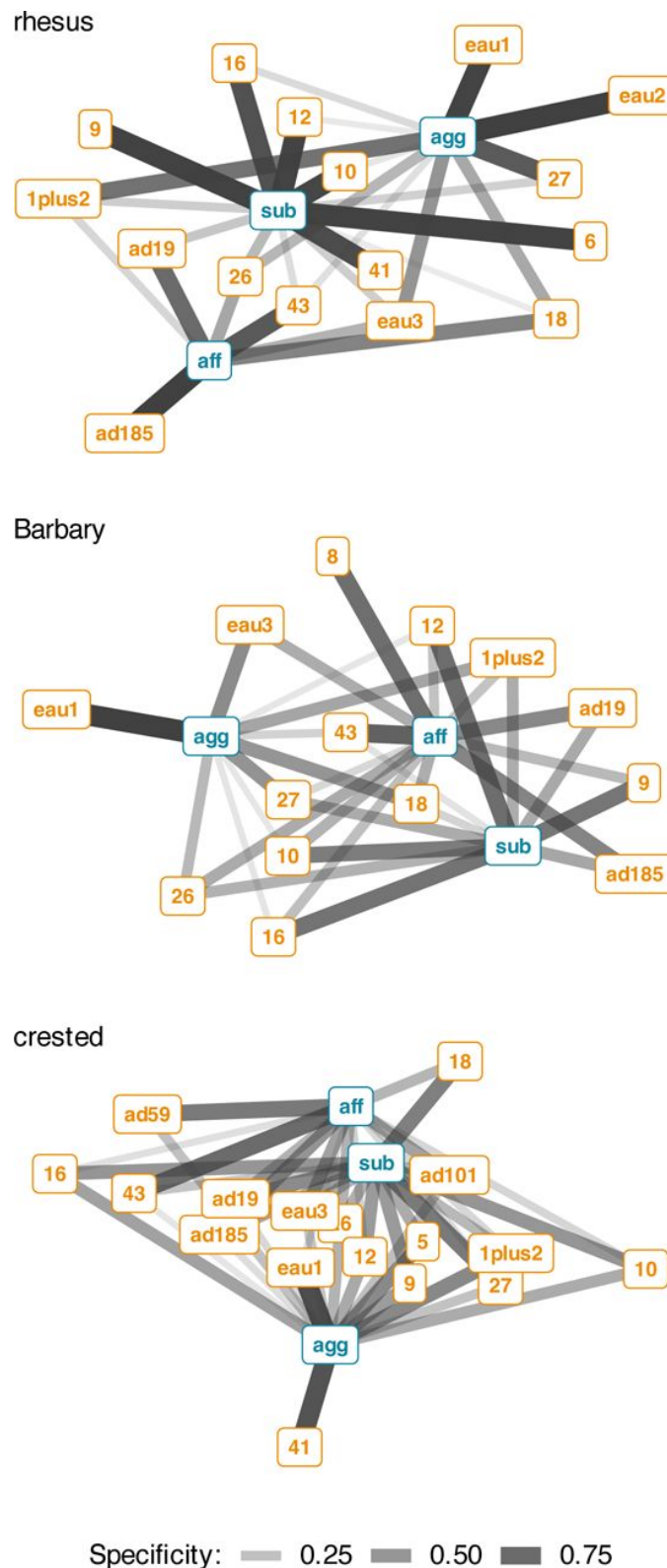


Figure 2

Bipartite network of single Action Units (orange) and social context (blue) for three species of macaques. Edges are shown for Action Units that occurred in at least 1% of observations per context. Edge thickness and transparency are weighted by specificity, which ranges from 0 (indicating an Action Unit is never observed in a context) to 1 (indicating an Action Unit is only observed in one context). Context abbreviations: agg = aggressive, aff = affiliative, sub = submissive.

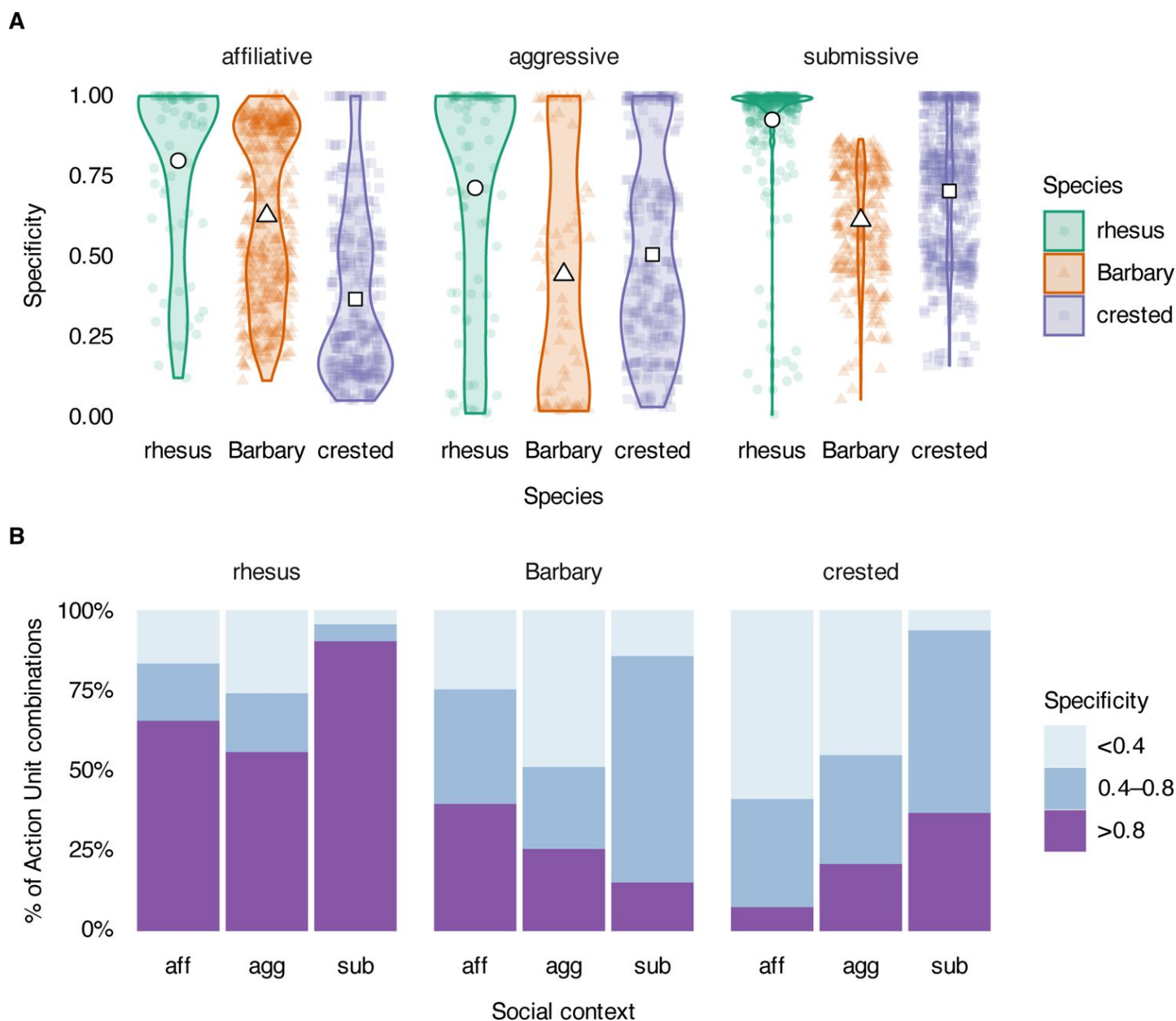


Figure 3

Specificity of Action Unit combinations that were used in at least 1% of observations per species per social context. Specificity ranges from 0 (indicating an Action Unit is never observed in a context) to 1 (indicating an Action Unit is only observed in one context). **(A)** Distribution of Action Unit combination specificity. Width of violin plots indicate the relative density of the data. Colored symbols indicate unique Action Unit combinations. White symbols indicate mean specificity. **(B)** Proportion of Action Unit combinations used with high (>0.8), moderate (0.4–0.8) or low (<0.4) specificity. Context abbreviations: agg = aggressive, aff = affiliative, sub = submissive.

and least accurate for crested macaques ($\kappa = 0.49$). The confusion matrices for model predictions are shown in table S1.

Discussion

We investigated the hypothesis that complex societies require more complex communication systems [1] by comparing the complexity of facial behavior of three species of macaques that vary in their degree of social tolerance and complexity. We defined facial behavior by the unique combinations of muscle movements visible in the face. Doing so allows for a much more precise description of facial behavior and captures subtle differences that are lost if facial expressions are classified as discrete categories. We quantified communicative complexity using three measures of uncertainty and predictability: entropy, context specificity, and prediction error. Collectively, our results suggest that the complexity of facial behavior is higher in species with a more tolerant—and therefore more complex—social style; complexity was highest for crested, followed by Barbary, and lowest in rhesus macaques. In light of what we know about the differences between macaque social systems, our results support the predictions of the social complexity hypothesis for communicative complexity.

The entropy ratio of facial behavior was highest in crested compared to Barbary and rhesus macaques, both overall and within each social context (affiliative, aggressive, submissive). This result suggests that crested macaques use a higher diversity of facial signals within each social context more frequently, resulting in the higher relative uncertainty in their use of facial behavior. Information theory defines information as the reduction in uncertainty once an outcome is learned [18]. By this definition, our data suggest that the facial behavior of crested macaques has the *potential* to communicate more information, compared to Barbary and rhesus macaques, although this would need to be explicitly tested in future studies. Our findings are in line with predictions of the social complexity hypothesis [1] given the differences in social styles between tolerant and intolerant macaques. In tolerant macaque societies, social interactions are less constrained by dominance [22] such that rates of counter aggression and reconciliation post-conflict are higher [25,30]. Thus, there is a greater variability in the kind of interactions that individuals have, potentially requiring the use of more diverse facial behavior to achieve social goals, particularly during conflicts. Similarly, strongly bonded chimpanzee (*Pan troglodytes*) dyads exhibit a larger repertoire of gestural communication than non-bonded dyads, presumably due to the former having more varied types of social interactions [31].

The overall entropy ratio of rhesus and Barbary macaques was similar, suggesting that they have similar communicative capacity using facial behavior. However, the entropy ratio differed when compared within social contexts; while relative entropy was higher for Barbary macaques in affiliative and submissive contexts, it was higher for rhesus macaques in aggressive contexts. One possible explanation may be due to the use of stereotyped signals of submission and dominance in each species. For example, subordinate rhesus macaques regularly exhibit stereotyped signals of submission (silent-bared-teeth), whereas dominant Barbary macaques regularly exhibit stereotyped threats (round-open-mouth) [26,27]. Frequent use of a stereotyped signal within a context reduces the overall diversity of signals, resulting in a lower entropy ratio for submission and aggression in rhesus and Barbary macaques, respectively. It has been suggested that in societies with high power asymmetries between individuals, such as in rhesus macaques, spontaneous signals of submission serve to prevent conflicts from escalating as well as increasing the tolerance of dominant individuals toward subordinates [27]. In societies with more moderate power asymmetries, such as in Barbary macaques, subordinates may be less motivated to spontaneously submit and thus dominants may need to assert their dominance with formalized threats more frequently [27].

While the entropy ratio captures the uncertainty of facial behavior used within a social context, context specificity captures the uncertainty generated when the same facial behavior is used flexibly across different social contexts. Overall, the context specificity of facial behavior was higher for the intolerant rhesus macaques as compared to the more tolerant Barbary and crested macaques across all three social contexts. This pattern occurred for both the mean specificity values and the proportion of Action Unit combinations used that had high (>0.8) specificity. Similarly, a previous study demonstrated that vocal calls of tolerant macaques are less context specific than in intolerant macaques [32]. There was not a clear difference in specificity between Barbary and crested macaques; specificity was higher for Barbary macaques in affiliative contexts, similar for both species in aggressive contexts, and higher for crested macaques in submissive contexts. These differences in context specificity of communicative signals across macaque species may be related to differences in power asymmetry in their respective societies, particularly as it relates to the risk of injury. For macaques, bites are far more likely to injure opponents than other types of contact aggression (e.g., grab, slap) and thus provide the best proxy for risk of injury [21]. The percentage of conflicts involving bites is much higher in the less tolerant rhesus macaque, compared to the more tolerant Barbary and crested macaques who have similar low rates of aggression involving bites [25,33]. Risky situations may promote the evolution of more conspicuous, stereotypical signals to reduce ambiguity [34]. Indeed, intolerant macaques such as the rhesus more commonly use formal signals of submission [26,27]. In our study, rhesus macaques used facial behavior with high specificity across all contexts but particularly in submissive contexts. If the same facial behavior (or signal in general) is used in multiple social contexts, its meaning may be uncertain and must be deduced from additional contextual cues [35]. When facial behavior is highly context specific, there is less uncertainty about the meaning of the signal and/or intention of the signaler. In a society where the risk of injury from aggression is high, it may be adaptive for individuals to use signals that are highly context specific or ritualized to reduce uncertainty about its meaning. By contrast, the lower risk of injury in Barbary and crested macaques may allow room for a greater variety of more nuanced behaviors during conflicts as well as higher rates of reconciliation post conflict [25,30].

In all three species of macaques, at least some facial muscle movements had low specificity and were therefore used across multiple social contexts that likely differed in valence. This finding is in line with the idea that communicative signals in primates are better interpreted as the signaler announcing its intentions and likely future behavior [36,37], and not necessarily as an expression of emotional state [28,29,36,38].

We found that a random forest classifier was least accurate at predicting social context from facial behavior for crested, followed by Barbary, and then rhesus macaques. The behavior of complex systems is generally harder to predict than simpler ones [16,17]. Thus, the relatively poorer performance of the classifier in crested macaques suggests that they have the most complex facial behavior. Nevertheless, the classifier was able to predict social context from facial behavior with better accuracy than expected by chance alone for all three species of macaque, including the crested. This result confirms the assumption that facial behavior in macaques is not used randomly and most likely has some communicative or predictive value [39]. It is worthwhile to reiterate here that completely random (and thus unpredictable) systems are not considered complex [19]. Therefore, the species with the highest entropy values, or unpredictability, could be interpreted as having a simpler communication system than a species with a moderately high entropy value or unpredictability. But the communications systems of living organisms are unlikely to be observed as random, otherwise they would not have evolved as signals. Therefore, working under the assumption that animal communication systems cannot possibly be random, we can conclude that the species whose communication system has the highest relative entropy and unpredictability is in fact the most complex [14].

In addition to social complexity, it is possible that other factors are related to the complexity of facial behavior. For example, primates with a larger body size have greater facial mobility [13, 40], which could allow for greater complexity of facial behavior. However, differences in mean body mass across the three macaques species of this study are small (rhesus: 6.5 kg; Barbary: 11.5 kg; crested: 7.4 kg) [41] with substantial overlap in body weight across adult individuals of the different species [42], and so it is unlikely to explain the differences in the complexity of facial behavior that we report in this study. The degree of terrestriality could also influence the evolution of facial signals due to more limited visibility in the canopy. However, differences in facial mobility across terrestrial and non-terrestrial primates are not significant once body size is controlled for [13]. Furthermore, all three species included in this study have comparable levels of terrestriality, spending the majority (52-72%) of the time on the ground [43–45]. Spatial spread and predation pressure could potentially also influence the use of facial signals. For example, when group spread is higher, reliance on facial signals could be lower since it is harder to perceive facial signals from a large distance. When predation pressure is higher, reliance on facial signals could be higher than, for example vocal signals, such as to not draw attention of predators to the signaler. There are currently no reliable data on predation pressure and spatial spread of the three species in their natural habitat but it could be a good avenue for future studies. It is also important to note that our study is correlational in nature and we cannot determine the direction of the link between social and communicative complexity. It is possible that an increase in communicative complexity evolved first, which then allowed for the evolution of more complex social systems. Finally, effectively, our comparison is limited to three species which is a small sample. However, the methodology we used is applicable to any species for which FACS is available (including other non-human primates, dogs, and horses; [11]), and therefore, we hope that other datasets will complement ours in the future.

Our results on the complexity of facial behavior in macaques is mirrored by previous studies showing that the complexity of vocal calls is similarly higher in tolerant compared to intolerant macaques [32, 46]. Although not all macaque facial expressions have a vocal component, vocalizations are fundamentally multisensory with both auditory and visual components, where different facial muscle contractions are partly responsible for different-sounding vocalizations [47]. Indeed, some areas of the brain in primates integrate visual and auditory information resulting in behavioral benefits [48]. For example, macaques detect vocalizations in a noisy environment faster when mouth movements are also visible, where faster reaction times are associated with a reduced latency in auditory cortical spiking activity [49]. Combined, these findings suggest that the evolution in the complexity of vocal and facial signals in macaques may be linked and the same may be true of primates in general. For instance, humans not only have the most complex calls (language) and gestures, but most likely use the most complex facial behavior as well, given that their general facial mobility is highest among primates (most Action Units) [12, 50]. In lemurs (Lemuriformes), the repertoire size of vocal, visual, and olfactory signals positively correlate with group size and each other, suggesting that complexity in all three communicative modalities coevolved with social complexity [51]. While the complexity of different communication modalities is likely interlinked and correlated with each other, future studies would ideally integrate signals from all modalities into a single communicative repertoire for each species. While collecting and analyzing data on multiple modalities of communication has historically been a challenge, such endeavors would be an important next step in the study of animal communication [52]. By breaking down signaling units to their smallest components, as we have done for facial behavior in this study, we may be able to define a “signal” by temporal co-activation of visual, auditory, and perhaps even olfactory cues, which would provide the most comprehensive picture of animal communication.

Study subjects and data collection

Behavioral data and video recordings were collected on one adult male and 31 adult female rhesus macaques (*M. mulatta*), on 18 adult male and 28 adult female Barbary macaques (*M. sylvanus*), and 17 adult male and 21 adult female crested macaques (*M. nigra*). See supplementary text for further details.

For all study groups and subjects, focal animal observations [53] lasting 15–30 minutes were conducted throughout the day in a pseudo-randomized order such that the number of days and time of day that each individual was observed was balanced. Videos of social interactions were recorded with a recording camera (Panasonic HDC-SD700, Bracknell, UK) during focal animal observations as well as *ad libitum*. Social behavior, including grooming, body contact, and agonistic interactions were recorded using a handheld smartphone or tablet with purpose-built software (rhesus: Animal Behavior Pro [54]; Barbary: CyberTracker (<http://cybertracker.org>), crested: Microsoft Excel).

Facial behavior and social context coding

Facial behavior was coded at the level of observable individual muscle movements using the Facial Action Coding System (FACS) [12], adapted for each species of macaque (MaqFACS): rhesus [55], Barbary [56], crested [10]. In FACS, individual observable muscle contractions are coded as unique Action Units (AU; e.g., upper lip raiser AU10). Some common facial movements where the underlying muscle is unknown are coded as Action Descriptors (AD; e.g., jaw thrust AD29). In MaqFACS, the lip-pucker AU18 has two subtle variations normally denoted as AU18i and AU18ii [55,56]. However, it was often difficult to reliably distinguish between these two subtle variations when coding videos, and so the lip-pucker was simply coded as AU18. We added a new Action Descriptor 185 (AD185) called jaw-oscillation, to denote the stereotyped movement of the jaw up and down. When combined with existing Action Units of lip movements, the jaw-oscillation AD185 allows for a more detailed and accurate coding of some facial behaviors that would otherwise be labeled as lipsmack (AD181), teeth-chatter, or jaw-wobble [10,55]. A complete list of Action Units and Action Descriptors coded in this study is given in table S2.

We coded facial behavior of adult individuals but included their interactions with any other group member regardless of age or sex. Each social interaction was labeled with a context; aggressive, submissive, affiliative, or unclear. We did not consider interactions in a sexual context because data for the rhesus macaques were only collected during the non-mating season. Social context was labeled from the point of view of the signaler based on their general behavior and body language (but not the facial behavior itself), during or immediately following the facial behavior. An aggressive context was considered when the signaler lunged or leaned forward with the body or head, charged, chased, or physically hit the interaction partner. A submissive context was considered when the signaler leaned back with the body or head, moved away, or fled from the interaction partner. An affiliative context was considered when the signaler approached another individual without aggression (as defined previously) and remained in proximity, in relaxed body contact, or groomed either during or immediately after the facial behavior. In cases where the behavior of the signaler did not match our context definitions, or displayed behaviors belonging to multiple contexts, we labeled the social context as unclear. Social context was determined from the video itself and/or from the matching focal behavioral data, if available. Videos were FACS coded frame-by-frame using the software BORIS [57] by AVR (rhesus, Barbary, crested), CP

(Barbary) and PRC (crested), who are certified FACS and MaqFACS coders. Inter-observer reliability was determined with the same index of agreement used by [12] for FACS, with the formula:

$$\frac{2(\text{The number of AUs on which both coders agreed})}{\text{Total number of AUs scored by both coders}}$$

An agreement rating of >0.7 was considered good [12] and was necessary for obtaining certification. To obtain a MaqFACS coding certification, AVR, CP, and PRC coded 23 video clips of rhesus macaques and the MaqFACS codes were compared to the data of other certified coders (<https://animalfacs.com>). The mean agreement ratings obtained were 0.85, 0.73, 0.83 for AVR, CP, and PRC, respectively. In addition, AVR and CP coded 7 videos of Barbary macaques with a mean agreement rating of 0.79. AVR and PRC coded 10 videos of crested macaques with a mean agreement rating of 0.74.

Table 1 shows the number of social interactions per species and context from which FACS codes were made.

Statistical analyses

Prior to analyses, MaqFACS data were formatted as a binary matrix with Action Units and Action Descriptors (hereafter simply Action Units) in the columns. Each row denoted an observation time block of 500ms, where if an Action Unit was active during this time block, it was coded 1 and coded 0 if not. Thus, each row contained information on the combination of facial muscle movements that were co-activated within a 500ms time window (**Table 1**). All 500ms time blocks per interaction were used in the statistical analyses in order to retain all the variation and complexity of the facial behavior (Action Unit combinations) used by the macaques. All statistical analyses were conducted in R (version 4.2.1) [58].

The observed entropy for each social context was calculated using Shannon's information entropy formula [18]:

$$H = - \sum_i^n p_i \log p_i$$

where n is the number of unique Action Unit combinations and p is the probability of observing each Action Unit combination in each social context. The expected maximum entropy was calculated by randomizing the data matrix while keeping the number of active Action Units per observation (row) the same. This process was repeated 100 times and the mean of the randomized entropy values was used as the expected entropy. Therefore, the expected entropy indicated the entropy of the system if facial muscle contractions occurred at random, while keeping the combination size of co-active muscle movements within the range observed in the data. The entropy ratio was calculated by dividing the observed entropy by the expected (maximum) entropy. To determine whether the entropy ratios for each species differed within social context, the entropy ratio was calculated on 100 bootstrapped samples of the data, resulting in a distribution of possible entropy ratios. If the distribution of bootstrapped entropy ratios did not overlap, the differences between entropy ratios were considered to be meaningful.

Species	Context	N interactions	N subjects	Duration (seconds)
rhesus	affiliative	193	29	1,197
	aggressive	413	32	2,050
	submissive	318	31	1,262
	unclear	121	30	802
Barbary	affiliative	683	43	4,897
	aggressive	585	44	2,128
	submissive	529	34	1,890
	unclear	603	45	3,500
crested	affiliative	241	35	1,918
	aggressive	62	23	284
	submissive	25	18	115
	unclear	107	25	684

Table 1

Total number of social interactions per species and social context that were MaqFACS coded. Note that combination of Action Units were grouped by time blocks of 500ms. Therefore, the number of observations in the data is twice the duration of the social interaction in seconds.

We calculated the specificity with which Action Unit combinations are associated with a social context within each species using the function “specificity” from the R package “NetFACS” (version 0.5.0) [59]. Due to an imbalanced number of observations across social contexts, contexts with fewer observations were randomly upsampled prior to the specificity calculation. During the upsampling procedure all observations of the minority contexts were kept, and new observations were randomly sampled to match the number of observations in the majority context. This procedure corrects for any bias in the specificity results from an imbalanced dataset (see fig. S1). Specificity is the conditional probability of a social context given that an Action Unit combination is observed, and ranges from 0 (when an Action Unit combination is never observed in a context) to 1 (when an Action Unit is only observed in one context). Low specificity values indicate that Action Units were used flexibly across multiple contexts whereas high values indicate that Action Units were used primarily in a single context. Specificity was calculated for all Action Unit combination sizes ranging from 1 to 11 (the maximum observed combination size) co-active Action Units. When reporting context specificity results, we excluded Action Unit combinations that occurred in less than 1% of observations within a social context because extremely rare signals do not impact the predictability of a communication system regardless of whether specificity is low or high. Therefore, excluding rare Action Unit combinations removes noise from the specificity results. We report the mean specificity of Action Unit combinations per social context and the proportion of Action Unit combinations that have high, moderate, or low specificity. For single Action Units we plotted bipartite networks that show how Action Units are connected to social context weighted by their specificity.

To predict social context from the combination of Action Units we fit a random forest classifier using the “tidymodels” R package (version 1.0.0) [60] using the function “ran_forest” with the engine set to “ranger” [61], 500 trees, 4 predictor columns randomly sampled at each split, and 10 as the minimum number of data points in a node required for splitting further. The data were randomly split into a training set (70%) and a test set (30%), while keeping the proportion of observations per social context the same in the training and test sets. Due to an imbalanced number of observations across social contexts, contexts with fewer observations were over-sampled in the training set using the SMOTE algorithm [62] to improve the classifier predictions. To assess the classifier performance, we report the kappa statistic, which denotes the observed accuracy corrected for the expected accuracy [63]. Kappa is 0 when the classifier performs at chance level and 1 when it shows perfect classification. Kappa values between 0 and 1 indicate how much better the classifier performed than chance (e.g., kappa of 0.5 indicates the classifier was 50% better than chance). Kappa is a more reliable estimate of model performance than accuracy alone when the relative sample size for each context is imbalanced, as was the case with our data.

Acknowledgements

We thank the German Primate Center (DPZ) for permission to collect data on the rhesus macaques, Uwe Schönmann for logistical support, and Julia Ostner for being our host at the DPZ. We thank Matt Lowatt and Ellen Merz for permission to collect data on the Barbary macaques at Trentham Monkey Forest. We thank the Indonesian State Ministry of Research and Technology (RISTEK), the Directorate General of Forest Protection and Nature Conservation (PHKA) and the Department for the Conservation of Natural Resources (BKSDA), North Sulawesi, for permission to access groups of crested macaques in the Tangkoko-Batuangus Nature Reserve. We thank Christof Neumann for statistical advice. This work was funded by the Leverhulme Trust (RPG2018-334).

Ethics

This work adhered to the Guidelines for the treatment of animals in behavioral research and teaching [64] and was approved by the Animal Welfare and Ethical Review Body of the University of Portsmouth (AWERB, approval number: 919B). The AWERB uses UK Home Office guidelines on the Animals (Scientific Procedures) Act 1986 when assessing proposals and adheres to the regulations of the European Directive 2010/63/EU. The German Primate Center also complies with the European Directive 2010/63/EU, as well as with the provisions of the German Animal Welfare Act.

Data availability

The data and R code used for all statistical analysis is available on GitHub, <https://github.com/avrincon/macaque-facial-complexity>.

Author contributions

Conceptualization: AVR, BMW, JD, and JM. Methodology: AVR, BMW, JD, AM and JM. Software: AVR, and AM. Formal analysis, validation and visualization: AVR. Investigation and data curation: AVR, CP, and PRC. Funding acquisition, project administration and supervision: BMW, JD, and JM. Writing—original draft: AVR. Writing—review & editing: All authors.

Competing interests

The authors declare that they have no competing interests.

References

1. Freeberg TM, Dunbar RIM, Ord TJ (2012) **Social complexity as a proximate and ultimate factor in communicative complexity** *Phil. Trans. R. Soc. B* **367**:1785–1801 <https://doi.org/10.1098/rstb.2011.0213>
2. Peckre L, Kappeler PM, Fichtel C (2019) **Clarifying and expanding the social complexity hypothesis for communicative complexity** *Behav. Ecol. Sociobiol* **73** <https://doi.org/10.1007/s00265-018-2605-4>
3. Raviv L, Peckre LR, Boeckx C (2022) **What is simple is actually quite complex: A critical note on terminology in the domain of language and communication** *J. Comp. Psychol* **136**:215–220
4. Bergman TJ, Beehner JC (2015) **Measuring social complexity** *Anim. Behav* **103**:203–209 <https://doi.org/10.1016/j.anbehav.2015.02.018>
5. Aureli F, Schaffner CM, Schino G (2022) **Variation in communicative complexity in relation to social structure and organization in non-human primates** *Phil. Trans. R. Soc. B* **377** <https://doi.org/10.1098/rstb.2021.0306>
6. Aureli F, Schino G (2019) **Social complexity from within: How individuals experience the structure and organization of their groups** *Behav. Ecol. Sociobiol* **73** <https://doi.org/10.1007/s00265-018-2604-5>
7. Kappeler PM (2019) **A framework for studying social complexity** *Behav. Ecol. Sociobiol* **73** <https://doi.org/10.1007/s00265-018-2601-8>
8. Keenan S, Lemasson A, Zuberbühler K (2013) **Graded or discrete? A quantitative analysis of Campbell's monkey alarm calls** *Anim. Behav* **85**:109–118 <https://doi.org/10.1016/j.anbehav.2012.10.014>
9. Dobson SD (2012) **Coevolution of facial expression and social tolerance in macaques** *Am. J. Primatol* **74**:229–235 <https://doi.org/10.1002/ajp.21991>
10. Clark PR, Waller BM, Burrows AM, Julle-Danière E, Agil M, Engelhardt A, Micheletta J (2020) **Morphological variants of silent bared-teeth displays have different social interaction outcomes in crested macaques (*Macaca nigra*)** *Am. J. Phys. Anthropol* **173**:411–422 <https://doi.org/10.1002/ajpa.24129>
11. Waller BM, Julle-Danière E, Micheletta J (2020) **Measuring the evolution of facial 'expression' using multi-species FACS** *Neurosci. Biobehav. Rev* **113**:1–11 <https://doi.org/10.1016/j.neubiorev.2020.02.031>
12. Ekman P, Friesen WV, Hager JC (2002) **Facial Action Coding System** *Salt Lake City: Research Nexus. See*
13. Dobson SD (2009) **Socioecological correlates of facial mobility in nonhuman anthropoids** *Am. J. Phys. Anthropol* **139**:413–420 <https://doi.org/10.1002/ajpa.21007>

14. Rebout N, Lone J-C, De Marco A, Cozzolino R, Lemasson A, Thierry B. (2021) **Measuring complexity in organisms and organizations** *R. Soc. Open. Sci* **8** <https://doi.org/10.1098/rsos.200895>
15. Sambrook T, Whiten A (1997) **On the nature of complexity in cognitive and behavioural science** *Theory Psychol* **7**:191–213 <https://doi.org/10.1177/0959354397072004>
16. McDaniel RR, Driebe DJ (2005) **Uncertainty and surprise in complex systems**. Springer Berlin, Heidelberg
17. Schuster P (2016) **How complexity originates: Examples from history reveal additional roots to complexity** *Complexity* **21**:7–12 <https://doi.org/10.1002/cplx.21841>
18. Shannon CE (1948) **A mathematical theory of communication** *The Bell System Technical Journal* **27**:379–423 <https://doi.org/10.1002/j.1538-7305.1948.tb01338.x>
19. Adami C (2002) **What is complexity?** *BioEssays* **24**:1085–1094 <https://doi.org/10.1002/bies.10192>
20. Thierry B (2007) **Unity in diversity: Lessons from macaque societies** *Evol. Anthropol* **16**:224–238 <https://doi.org/10.1002/evan.20147>
21. Thierry B (2021) **Where do we stand with the covariation framework in primate societies?** *Am. J. Biol. Anthropol* **128**:5–25 <https://doi.org/10.1002/ajpa.24441>
22. Balasubramaniam KN, et al. (2012) **Hierarchical steepness, counter-aggression, and macaque social style scale** *Am. J. Primatol* **74**:915–925 <https://doi.org/10.1002/ajp.22044>
23. Sueur C, Petit O, De Marco A, Jacobs AT, Watanabe K, Thierry B. (2011) **A comparative network analysis of social style in macaques** *Anim. Behav* **82**:845–852 <https://doi.org/10.1016/j.anbehav.2011.07.020>
24. Berman C, Thierry B (2010) **Variation in kin bias: Species differences and time constraints in macaques** *Behaviour* **147**:1863–1887 <https://doi.org/10.1163/000579510X539691>
25. Duboscq J, Micheletta J, Agil M, Hodges K, Thierry B, Engelhardt A (2013) **Social tolerance in wild female crested macaques (Macaca nigra) in Tangkoko-Batuangus nature reserve, Sulawesi, Indonesia** *Am. J. Primatol* **75**:361–375 <https://doi.org/10.1002/ajp.22114>
26. de Waal FBM, Luttrell LM. (1985) **The formal hierarchy of rhesus macaques: An investigation of the bared-teeth display** *Am. J. Primatol* **9**:73–85 <https://doi.org/10.1002/ajp.1350090202>
27. Preuschoft S, Schaik C. (2000) **Dominance and communication: Conflict management in various social settings**. In *Natural Conflict Resolution* (eds F Aureli, FBM De Waal)
28. Waller BM, Whitehouse J, Micheletta J (2017) **Rethinking primate facial expression: A predictive framework** *Neurosci. Biobehav. Rev* **82**:13–21 <https://doi.org/10.1016/j.neubiorev.2016.09.005>
29. Fridlund AJ (1994) **Human facial expression: An evolutionary view**

30. Thierry B, Aureli F, Nunn CL, Petit O, Abegg C, de Waal FBM. (2008) **A comparative study of conflict resolution in macaques: Insights into the nature of trait covariation** *Anim. Behav* **75**:847–860 <https://doi.org/10.1016/j.anbehav.2007.07.006>
31. Amici F, Liebal K (2022) **The social dynamics of complex gestural communication in great and lesser apes (*Pan troglodytes*, *Pongo abelii* *Symphalangus syndactylus*)**. *Phil. Trans. R. Soc. B* **377** <https://doi.org/10.1098/rstb.2021.0299>
32. Rebout N, et al. (2022) **Tolerant and intolerant macaques differ in the context specificity of their calls and how they ‘comment’ on the interactions of others** *Behav. Ecol. Sociobiol* **76** <https://doi.org/10.1007/s00265-022-03177-7>
33. Tyrrell M, Berman CM, Duboscq J, Agil M, Sutrisno T, Engelhardt A (2020) **Avoidant social style among wild crested macaque males (*Macaca nigra*) in Tangkoko nature reserve, Sulawesi, Indonesia** *Behaviour* **157**:451–491 <https://doi.org/10.1163/1568539X-bja10009>
34. Clark PR, Waller BM, Agil M, Micheletta J (2022) **Crested macaque facial movements are more intense and stereotyped in potentially risky social interactions** *Phil. Trans. R. Soc. B* **377** <https://doi.org/10.1098/rstb.2021.0307>
35. Seyfarth RM, Cheney DL (2017) **The origin of meaning in animal signals** *Anim. Behav* **124**:339–346 <https://doi.org/10.1016/j.anbehav.2016.05.020>
36. Cheney DL, Seyfarth RM (2018) **Flexible usage and social function in primate vocalizations** *Proc. Natl. Acad. Sci* **115**:1974–1979 <https://doi.org/10.1073/pnas.1717572115>
37. Fischer J, Price T (2017) **Meaning, intention, and inference in primate vocal communication** *Neurosci. Biobehav. Rev* **82**:22–31 <https://doi.org/10.1016/j.neubiorev.2016.10.014>
38. Barrett LF, Adolphs R, Marsella S, Martinez AM, Pollak SD (2019) **Emotional expressions reconsidered: Challenges to inferring emotion from human facial movements** *Psychol. Sci. Public Interest* **20**:1–68 <https://doi.org/10.1177/1529100619832930>
39. Waller BM, Whitehouse J, Micheletta J (2016) **Macaques can predict social outcomes from facial expressions** *Anim. Cogn* **19**:1031–1036 <https://doi.org/10.1007/s10071-016-0992-3>
40. Santana SE, Dobson SD, Diogo R (2014) **Plain faces are more expressive: Comparative study of facial colour, mobility and musculature in primates** *Biol. Lett* **10** <https://doi.org/10.1098/rsbl.2014.0275>
41. Jones KE, et al. (2009) **PanTHERIA: A species-level database of life history, ecology, and geography of extant and recently extinct mammals** *Ecology* **90**:2648–2648 <https://doi.org/10.1890/08-1494.1>
42. Smith RJ, Jungers WL (1997) **Body mass in comparative primatology** *J. Hum. Evol* **32**:523–559 <https://doi.org/10.1006/jhev.1996.0122>
43. Khatiwada S, Paudel PK, Chalise MK, Ogawa H (2020) **Comparative ecological and behavioral study of *Macaca assamensis* and *M. mulatta* in Shivapuri Nagarjun National Park, Nepal** *Primates* **61**:603–621 <https://doi.org/10.1007/s10329-020-00810-9>
44. O’Brien TG, Kinnaird MF (1997) **Behavior, diet, and movements of the Sulawesi crested black macaque (*Macaca nigra*)** *Int. J. Primatol* **18**:321–351 <https://doi.org/10.1023/A:1026330332061>

45. Alami AE, Chait A (2014) **Comparison of the terrestriality of Barbary macaques (*Macaca sylvanus*) between tourist and wild sites in the region of Ouzoud, Morocco** *Mammalia* **78**:539–542 <https://doi.org/10.1515/mammalia-2013-0061>
46. Rebout N, et al. (2020) **Tolerant and intolerant macaques show different levels of structural complexity in their vocal communication** *Proc. R. Soc. B* **287** <https://doi.org/10.1098/rspb.2020.0439>
47. Ghazanfar AA, Takahashi DY (2014) **The evolution of speech: Vision, rhythm, cooperation** *Trends Cogn. Sci* **18**:543–553 <https://doi.org/10.1016/j.tics.2014.06.004>
48. Ghazanfar AA, Eliades SJ (2014) **The neurobiology of primate vocal communication** *Curr. Opin. Neurobiol* **28**:128–135 <https://doi.org/10.1016/j.conb.2014.06.015>
49. Chandrasekaran C, Lemus L, Ghazanfar AA (2013) **Dynamic faces speed up the onset of auditory cortical spiking responses during vocal detection** *Proc. Natl. Acad. Sci* **110**:E4668–E4677 <https://doi.org/10.1073/pnas.1312518110>
50. Dobson SD (2009) **Allometry of facial mobility in anthropoid primates: Implications for the evolution of facial expression** *Am. J. Phys. Anthropol* **138**:70–81 <https://doi.org/10.1002/ajpa.20902>
51. Fichtel C, Kappeler PM (2022) **Coevolution of social and communicative complexity in lemurs** *Phil. Trans. R. Soc. B* **377** <https://doi.org/10.1098/rstb.2021.0297>
52. Liebal K, Slocombe KE, Waller BM (2022) **The language void 10 years on: Multimodal primate communication research is still uncommon** *Ethol. Ecol. Evol* **34**:274–287 <https://doi.org/10.1080/03949370.2021.2015453>
53. Altmann J (1974) **Observational study of behavior: Sampling methods** *Behaviour* **49**:227–266 <https://doi.org/10.1163/156853974X00534>
54. Newton-Fisher NE. (2020) **Newton-Fisher NE. 2020 Animal behaviour pro. Mobile App, version 1.5. See <https://itunes.apple.com/gb/app/animal-behaviour-pro/id579588319?mt=8>. Animal behaviour pro. Mobile App, version 1.5**
55. Parr LA, Waller BM, Burrows AM, Gothard KM, Vick SJ (2010) **Brief communication: MaqFACS: A muscle-based facial movement coding system for the rhesus macaque** *Am. J. Phys. Anthropol* **143**:625–630 <https://doi.org/10.1002/ajpa.21401>
56. Julle-Danière É, Micheletta J, Whitehouse J, Joly M, Gass C, Burrows AM, Waller BM (2015) **MaqFACS (Macaque Facial Action Coding System) can be used to document facial movements in Barbary macaques (*Macaca sylvanus*)** *PeerJ* **3** <https://doi.org/10.7717/peerj.1248>
57. Friard O, Gamba M (2016) **BORIS: A free, versatile open-source event-logging software for video/audio coding and live observations** *Methods Ecol. Evol* **7**:1325–1330 <https://doi.org/10.1111/2041-210X.12584>
58. R Core Team (2022) **R: A language and environment for statistical computing**
59. Mielke A, Waller BM, Pérez C, Rincon AV, Duboscq J, Micheletta J (2022) **NetFACS: Using network science to understand facial communication systems** *Behav. Res. Methods* **54**:1912–1927 <https://doi.org/10.3758/s13428-021-01692-5>

60. Kuhn M, Wickham H. (2020) **Tidymodels: A collection of packages for modeling and machine learning using tidyverse principles**
61. Wright MN, Ziegler A (2017) **Ranger: A fast implementation of random forests for high dimensional data in C++ and R**. *J. Stat. Softw* **77**:1–17 <https://doi.org/10.18637/jss.v077.i01>
62. Chawla NV, Bowyer KW, Hall LO, Kegelmeyer WP (2002) **SMOTE: Synthetic minority over-sampling technique**. *J. Artif. Intell. Res* **16**:321–357 <https://doi.org/10.1613/jair.953>
63. Cohen J (1960) **A coefficient of agreement for nominal scales**. *Educ. Psychol. Meas* **20**:37–46 <https://doi.org/10.1177/001316446002000104>
64. ASAB Ethical Committee, ABS Animal Care Committee (2022) **Guidelines for the treatment of animals in behavioural research and teaching**. *Anim. Behav* **183** [https://doi.org/10.1016/S0003-3472\(21\)00389-4](https://doi.org/10.1016/S0003-3472(21)00389-4)

Author information

Alan V. Rincon

Department of Psychology, Centre for Comparative and Evolutionary Psychology, University of Portsmouth, Portsmouth PO1 2DY, UK

For correspondence: avrincon1@gmail.com

ORCID iD: [0000-0001-6181-0152](https://orcid.org/0000-0001-6181-0152)

Bridget M. Waller

Centre for Interdisciplinary Research on Social Interaction, Department of Psychology, Nottingham Trent University, Nottingham NG1 4BU, UK

ORCID iD: [0000-0001-6303-7458](https://orcid.org/0000-0001-6303-7458)

Julie Duboscq

UMR7206 Eco-Anthropology, CNRS-MNHN-Université Paris Cité, Paris 75016, France

ORCID iD: [0000-0002-6342-6709](https://orcid.org/0000-0002-6342-6709)

Alexander Mielke

Origins of Mind Group, School of Psychology and Neuroscience, University of St Andrews, St Andrews KY16 9JP, UK

ORCID iD: [0000-0002-8847-6665](https://orcid.org/0000-0002-8847-6665)

Claire Pérez

Department of Psychology, Centre for Comparative and Evolutionary Psychology, University of Portsmouth, Portsmouth PO1 2DY, UK

Peter R. Clark

Department of Psychology, Centre for Comparative and Evolutionary Psychology, University of Portsmouth, Portsmouth PO1 2DY, UK, School of Psychology, University of Lincoln, Lincoln LN6 7TS, UK

ORCID iD: [0000-0001-6725-7781](https://orcid.org/0000-0001-6725-7781)

Jérôme Micheletta

Department of Psychology, Centre for Comparative and Evolutionary Psychology, University of Portsmouth, Portsmouth PO1 2DY, UK

For correspondence: jerome.micheletta@port.ac.uk

Editors

Reviewing Editor

Ammie Kalan

University of Victoria, Canada

Senior Editor

Detlef Weigel

Max Planck Institute for Biology Tübingen, Germany

Reviewer #1 (Public Review):

After revision, the manuscript is clearly improved and I thank the authors for their efforts. Yet, two contentious issues remain.

Firstly, I am skeptical whether the circularity issue has been resolved.

The authors equate uncertainty in the outcome of interactions with social complexity and they then diagnose for these three species that higher social complexity correlates with higher communicative complexity. Yet, there is still an inherent link between the occurrence of signals and other behaviours that allow the authors to determine the outcome of an interaction.

I do agree with the authors' conclusion that the three species vary in terms of the predictability of their signaling behaviour and the outcome of interactions. I just think the observed link between the two is not very surprising or informative, but rather inevitable.

Secondly, I am still not convinced that visual communication is more prevalent in situations with higher predation pressure. There are two reasons: relying on visual communication requires that the recipients, typically one's group members, are actually looking at the signaler when they produce the signal. The vocal-auditory channel in contrast, has a much higher potential to reach all recipients, even when visual communication is impaired. In addition, the idea that predators use acoustic signals to single out individuals and preferentially attack them, is poorly corroborated by data, especially for terrestrial predators. In contrast, there is ample evidence that prey species direct their calls at terrestrial predators (mobbing calls against snakes, antelope vigorously snorting against lions and leopards). See also this paper by Griesser (PMID 23941356).

Reviewer #2 (Public Review):

This is a well-written manuscript about a strong comparative study of diversity of facial movements in three macaque species to test arguments about social complexity influencing communicative complexity.

Author Response

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

This study investigates the context-specificity of facial expressions in three species of macaques to test predictions for the 'social complexity hypothesis for communicative complexity'. This hypothesis has garnered much attention in recent years. A proper test of this hypothesis requires clear definitions of 'communicative complexity' and 'social complexity'. Importantly, these two facets of a society must not be derived from the same data because otherwise, any link between the two would be trivial. For instance, if social complexity is derived from the types of interactions individuals have, and different types of signals accompany these interactions, we would not learn anything from a correlation between social and communicative complexity, as both stem from the same data.

The authors of the present paper make a big step forward in operationalising communicative complexity. They used the Facial Action Coding System to code a large number of facial expressions in macaques. This system allows decomposing facial expressions into different action units, such as 'upper lid raiser', 'upper lip raiser' etc.; these units are closely linked to activating specific muscles or muscle groups. Based on these data, the authors calculated three measures derived from information theory: entropy, specificity and prediction error. These parts of the analysis will be useful for future studies.

The three species of macaque varied in these three dimensions. In terms of entropy, there were differences with regard to context (and if there are these context-specific differences, then why pool the data?). Barbary and Tonkean macaques showed lower specificity than rhesus macaques. Regarding predicting context from the facial signals, a random forest classifier yielded the highest prediction values for rhesus monkeys. These results align with an earlier study by Preuschoft and van Schaik (2000), who found that less despotic species have greater variability in facial expressions and usage.

*Crucially, the three species under study are also known to vary in terms of their social tolerance. According to the highly influential framework proposed by Bernard Thierry, the members of the genus *Macaca* fall along a graded continuum from despotic (grade 1) to highly tolerant (grade 4). The three species chosen for the present study represent grade 1 (rhesus monkeys), grade 3 (Barbary macaques), and grade 4 (Tonkean macaques).*

The authors of the present paper define social complexity as equivalent to social tolerance - but how is social tolerance defined? Thierry used aggression and conflict resolution patterns to classify the different macaque species, with the steepness of the rank hierarchy and the degree of nepotism (kin bias) being essential. However, aggression and conflict resolution are accompanied by facial gestures. Thus, the authors are looking at two sides of the same coin when investigating the link between social complexity (as defined by the authors) and communicative complexity. Therefore, I am not convinced that this study makes a significant advance in testing the social complexity for communicative complexity hypothesis. A further weakness is that - despite the careful analysis - only three species were considered; thus, the effective sample size is very small.

Social tolerance in macaques is defined by various covarying traits, among which rates of counter-aggression and conflict resolution are only two of many included (see Thierry 2021 for a recent discussion and review). We do not deviate from Thierry's definition of social tolerance. We simply highlight that the constellation of behavioral traits in the most tolerant macaque species results in a social environment where the outcome of social interactions is more uncertain (see introduction lines 102-114). As we argue throughout the paper, higher uncertainty can be used as a proxy for higher complexity and thus we conclude that the most tolerant macaque species have the highest social complexity. While most social behavior in

macaques is accompanied by some facial behavior, we were careful to define social contexts only from the body language/behavior (e.g., lunge for aggression, grooming for affiliation) of the individuals involved and ignored the facial behavior used (see method lines 371-381). Therefore, the facial behavior of macaques (communication signals) was not used in defining either social tolerance (and by extension complexity) or the social context in which it was used. We feel like this appropriately minimizes any elements of circularity in the analysis of social and communicative complexity.

Regarding the effective sample size of three species, we agree that it is small, and it is a limitation of this study. However, the methodology we used is applicable to any species for which FACS is available (including other non-human primates, dogs, and horses), and therefore, we hope that other datasets will complement ours in the future. Nevertheless, we now acknowledge this limitation in the discussion (lines 314-317).

Reviewer #2 (Public Review):

This is a well-written manuscript about a strong comparative study of diversity of facial movements in three macaque species to test arguments about social complexity influencing communicative complexity. My major criticism has to do with the lack of any reporting of inter-observer reliability statistics - see comment below. Reporting high levels of inter-observer reliability is crucial for making clear the authors have minimized chances of possible observer biases in a study like this, where it is not possible to code the data blind with regard to comparison group. My other comments and questions follow by line number:

We agree that inter-observer coding reliability is an important piece of information. We now report in more detail the inter-observer reliability tests that we conducted on lines 384-392.

38-40. Whereas I am an advocate of this hypothesis and have tested it myself, the authors should probably comment here, or later in the discussion, about the reverse argument - greater communicative complexity (driven by other selection pressures) could make more complicated social structures possible. This latter view was the one advocated by McComb & Semple in their foundational 2005 Biology Letters comparative study of relationships between vocal repertoire size and typical group size in non-human primate species.

It is true that an increase in communicative complexity could allow/drive an increase in social complexity. Unfortunately our data is correlational in nature and we cannot determine the direction of causality. We added such a statement to the discussion (lines 311-314).

72-84 and 95-96. In the paragraph here, the authors outline an argument about increasing uncertainty / entropy mapping on to increasing complexity in a system (social or communicative). In lines 95-96, though, they fall back on the standard argument about complex systems having intermediate levels of uncertainty (complete uncertainty roughly = random and complete certainty roughly = simple). Various authors have put forward what I think are useful ways of thinking about complexity in groups - from the perspective of an insider (i.e., a group member, where greater randomness is, in fact, greater complexity) vs from the perspective of an outside (i.e., a researcher trying to quantify the complexity of the system where it is relatively easy to explain a completely predictable or completely random system but harder to do so for an intermediately ordered or random system). This sort of argument (Andrew Whiten had an early paper that made this argument) might be worth raising here or later in the discussion? (I'm also curious where the authors' sentiments lie for this question - they seem to touch on it in lines 285-287, but I think it's worth unpacking a little more here!)

In this study we used three measures of uncertainty (entropy, context specificity, and prediction error) to approximate complexity. However, maximum entropy or uncertainty would be achieved in a system that is completely random (and thus be considered simple). Therefore, the species with the highest entropy values, or unpredictability, could be interpreted as having a simpler communication system than a species with a moderately high entropy/unpredictability value. Our argument is that animal communication systems cannot possibly be random, otherwise they would not have evolved as signals. In systems where we know the highest entropy (or unpredictability) will not be due to randomness, as is the case with animal social interactions and communication, we can conclude that the system with the highest uncertainty is the most complex. We have now expanded upon this point in the discussion (lines 286-294). See also response to reviewer 1 below.

115-129. See also:

Maestriperi, D. (2005). "Gestural communication in three species of macaques (Macaca mulatta, M. nemestrina, M. arctoides): use of signals in relation to dominance and social context." Gesture 5: 57-73.

Maestriperi, D. and K. Wallen (1997). "Affiliative and submissive communication in rhesus macaques." Primates 38(2): 127-138.

On that note, it is probably worth discussing in this paragraph and probably later in the discussion exactly how this study differs from these earlier studies of Maestriperi. I think the fact that machine learning approaches had the most difficulty assigning crested data to context is an important methodological advance for addressing these sorts of questions - there are probably other important differences between the authors' study here and these older publications that are worth bringing up.

Our study differs from these two studies in that the studies above classified facial behavior into discrete categories (e.g., bared-teeth, lip-smack), whereas we adopted a bottom-up approach and made no a priori assumptions about which movements are relevant. We broke down facial behavior down to their individual muscle movements (i.e., Action Units). Measuring facial behavior at the level of individual muscle movements allows for a more detailed and objective description of the complexity of facial behavior. This is a general point in advancing the study of facial behavior that is discussed in the introduction (lines 60-71) and discussion (lines 206-208). The reason we don't draw a direct comparison with the studies above is because they had a slightly different focus. Our study was more focused on complexity of the (facial) communication system in general rather than comparing whether the different species use the same facial behavior in the same/different social contexts.

220-222. What is known about visual perception in these species? Recent arguments suggest that more socially complex species should have more sensitive perceptual processing abilities for other individuals' signals and cues (see Freeberg et al. 2019 Animal Behaviour). Are there any published empirical data to this effect, ideally from the visual domain but perhaps from any domain?

This is an interesting point. We are not aware of any studies showing differences in visual perceptions within the macaque genus. Both crested macaques and rhesus macaques are able to discriminate between individuals and facial expressions in match-to-sample tasks with comparable performances (Micheletta et al., 2015a, 2015b; Parr et al. 2008; Parr & Heinz, 2009). Similarly, several macaque species are sensitive to gaze shifts from conspecifics (Tomasello et al. 1998; Teufel et al. 2010; Micheletta & Waller, 2012).

274-277. I am not sure I follow this - could not different social and non-social contexts produce variation in different affective states such that "emotion"-based signals could be as flexible / uncertain as seemingly volitional / information-based / referential-like signals? This issue is probably too far away from the main points of this paper, but I suspect the authors' argument in this sentence is too simplified or overstated with regard to more affect-based signals.

Emotion-based signals could, in theory, also produce flexible signals and it is possible that some facial expressions reflect an emotional state. However, some previous studies have suggested that facial expressions are only used as a display of emotion, rather than such signals having evolved for a different function such as announcing future intentions. In our study we found that macaques used, in some cases, the same facial expressions (i.e. combination of Action Units) in at least two different social contexts that, presumably, differed in their emotional valence. Thus, it is unlikely that particular facial expressions are bound to a single emotion. We think that this is an important point to make even though it is slightly beyond the scope of our paper.

288 on. Given there are only three species in this study, the chances of one of the species being the 'most complex' in any measure is 0.33. Although I do not believe this argument I am making here, can the authors rule out the possibility that their findings related to crested macaques are all related to chance, statistically speaking?

We are not aware of a way to rule out this possibility. However, we believe that we are appropriately cautious throughout the paper and acknowledge that having only investigated three species is a limitation of this study in the discussion (lines 314-317, see also our response to reviewer 1 above).

329-330. The fact that only one male rhesus macaque was assessed here seems problematic, given the balance of sexes in the other two species. Can the authors comment more on this - are the gestures they are studying here identical across the sexes?

We agree it would have been preferable to collect data on more than one male rhesus macaque, but that was unfortunately not possible. We are not aware of any studies showing differences in the use of facial behavior between male and female rhesus macaques. If differences exist, most likely these would occur in a sexual/mating context. However, in our study we only considered affiliative (non-sexual), submissive, and aggressive contexts, where we have no a priori reason to believe that there are sex differences.

354-371. Inter-observer reliability statistics are required here - one of the authors who did not code the original data set, or a trained observer who is not an author, could easily code a subset of the video files to obtain inter-observer reliability data. This is important for ruling out potential unconscious observer biases in coding the data.

We agree this is an important piece of information. We now report in more detail the inter-observer reliability tests that we conducted on lines 384-392:

“An agreement rating of >0.7 was considered good [Ekman et al 2002] and was necessary for obtaining certification. To obtain a MaqFACS coding certification, AVR, CP, and PRC coded 23 video clips of rhesus macaques and the MaqFACS codes were compared to the data of other certified coders (<https://animalfacs.com>).

The mean agreement ratings obtained were 0.85, 0.73, 0.83 for AVR, CP, and PRC, respectively. In addition, AVR and CP coded 7 videos of Barbary macaques with a mean agreement rating

of 0.79. AVR and PRC coded 10 videos of crested macaques with a mean agreement rating of 0.74.”

Reviewer #1 (Recommendations For The Authors):

Given the long debate on the concept of information exchange in animal communication, I would also recommend being more careful with the term 'exchanges of information' (line 271). Perhaps it's better to be agnostic in the context of this paper.

As suggested, we now changed the phrasing to focus on the behavior of the animals, rather than suggesting that information is being exchanged (lines 270-273),

Line 281: "This result confirms the assumption that facial behaviour in macaques is not used randomly": the authors are knocking down a straw man. Nobody who has ever studied animal communication would consider that signals occur randomly. Otherwise, they would not have evolved as signals.

Indeed, nobody claims that animal communication signals are used randomly. Although it may be taken for granted, we feel it is worthwhile to reiterate this point, given that we used relative entropy and prediction error as measures of complexity. For instance, maximum entropy or unpredictability would be achieved in a system that is completely random (and thus be considered simple). Therefore, the species with the highest entropy values, or lowest predictability, could be interpreted as having a simpler communication system than a species with a moderately high entropy value. But if we are working under the assumption that animal communication systems cannot possibly be random, then we can conclude that the species whose communication system has the highest entropy is in fact the most complex. We tried to make this justification clearer in the discussion (lines 285-294).

I did not follow why there is a higher reliance on facial signals when predation pressure is higher. Apart from the fact that the authors cannot address this question, they may want to reconsider this idea altogether.

We now expand on the logic of why predation pressure might affect the use of facial signals (see lines 308-309): “When predation pressure is higher, reliance on facial signals could be higher than, for example vocal signals, such as to not draw attention of predators to the signaller.”

Technical comments:

One methodological issue that requires clarification is what the units of analysis are. The authors write that each row in their analysis denoted an observation time of 500 ms. How many rows did the authors assemble? The authors mention a sample size of > 3000 social interactions in the abstract. How did they define social interactions? And how many 'time windows' of 500 ms were obtained? Did they take one window per interaction or several? If several, then how was this move accounted for in the analysis? The reporting needs to be more accurate here. Most likely, the bootstrapping took care of biases in the data, but still, this information needs to be provided.

We have now added some additional information to the method section. Social interactions for each context had the following definitions: “Social context was labeled from the point of view of the signaler based on their general behavior and body language (but not the facial behavior itself), during or immediately following the facial behavior. An aggressive context was considered when the signaler lunged or leaned forward with the body or head, charged, chased, or physically hit the interaction partner. A submissive context was considered when

the signaler leaned back with the body or head, moved away, or fled from the interaction partner. An affiliative context was considered when the signaler approached another individual without aggression (as defined previously) and remained in proximity, in relaxed body contact, or groomed either during or immediately after the facial behavior. In cases where the behavior of the signaler did not match our context definitions, or displayed behaviors belonging to multiple contexts, we labeled the social context as unclear. Social context was determined from the video itself and/or from the matching focal behavioral data, if available.” (lines 371-382). The total duration of all social interactions per social context, and thus the number of 500ms windows/rows, have been added to Table 1 (lines 395-397). There were several 500ms windows per social interaction. All 500ms time blocks per interaction were used in the statistical analyses in order to retain all the variation and complexity of the facial behavior (Action Unit combinations) used by the macaques (lines 403-405). Indeed the bootstrapping procedure was used to account for any biases in the data.

Overall, I would recommend providing more information on the actual behaviour of the animals. The paper is strong in handling highly derived indices representing the behaviour, but the reader learns little about the animals' behaviour. Thus, it would be great if statements about the entropy ratio were translated into what these measures represent in real life. For context specificity, this is clear, but for entropy, not so much.

A high entropy ratio essentially suggests that a species uses a high variety of unique facial behavior/signals and all signals in the repertoire are used roughly equally often (rather than one facial behavior being used 90% of the time and others rarely used). We have tried our best to better explain this point in the introduction (lines 75-81) and discussion (lines 215-222). Discussing exactly what these signals are and what they mean was beyond the scope of this paper.

Line 106: nepotism, not kinship

Changed as suggested (line 106).

Line 113: I would avoid statements about how a monkey society is perceived by its members.

We think that noting how individuals may perceive their social environment is worthwhile when defining social complexity, so have retained this point but changed the phrasing to be more speculative (lines 112-113).

Line 329: I was very surprised that only one male was represented in the data for rhesus monkeys. The authors try to wriggle their way out of this issue in the supplementary material ("Therefore, we have no a priori reason to expect an overall difference in the diversity and complexity of facial behaviour between the sexes"), but I think this is a major shortcoming of the analysis. They should ascertain whether there are no sex differences in the other two species regarding their variables of interest. They could then make a very cautious case for there being no sex differences in rhesus either. But of course, they would not know for sure.

As with our response to reviewer 2 above, we agree that it would have been preferable to collect data on more than one male rhesus macaque, but that was unfortunately not possible. We are not aware of any studies showing differences in the use of facial behavior between male and female rhesus macaques. If differences exist, most likely these would occur in a sexual/mating context. However, in our study we only considered affiliative (non-sexual), submissive, and aggressive contexts, where we have no a priori reason to believe that there

are sex differences. Looking at sex differences in the use of facial behavior would be a worthwhile study on its own, but it is outside the scope of this paper.

This paper would make a stronger contribution if it focussed on the comparative analysis of facial expressions and removed the attempt of testing the social complexity for communicative complexity hypothesis.

A comparative analysis of the contextual use of specific facial movements is important. But this paper is focused on making a more general comparison of the communication style and complexity across species. The social complexity hypothesis for communicative complexity is one of the key theoretical frameworks for such an investigation and allows us to frame our study in a broader context. We contribute important data on 3 species with methods that can be replicated and extended to others species. Therefore, we believe that it is a worthy contribution to investigations of the evolution of complex communication.

REFERENCES

- Micheletta, J., J. Whitehouse, L.A. Parr, and B.M. Waller. 'Facial Expression Recognition in Crested Macaques (*Macaca nigra*)'. *Animal Cognition* 18 (2015): 985–90. <https://doi.org/10/f7fvnh>.
- Micheletta, Jérôme, Jamie Whitehouse, Lisa A. Parr, Paul Marshman, Antje Engelhardt, and Bridget M. Waller. 'Familiar and Unfamiliar Face Recognition in Crested Macaques (*Macaca nigra*)'. *Royal Society Open Science* 2 (2015): 150109. <https://doi.org/10/ggx9k9>.
- Parr, L. A., and M. Heintz. 'Facial Expression Recognition in Rhesus Monkeys, *Macaca mulatta*'. *Animal Behaviour* 77 (2009): 1507–13. <https://doi.org/10/bbsp5n>.
- Parr, L.A., M. Heintz, and G. Pradhan. 'Rhesus Monkeys (*Macaca mulatta*) Lack Expertise in Face Processing'. *Journal of Comparative Psychology* 122 (2008): 390–402. <https://doi.org/10/d7w6bv>.
- Micheletta, J., and B.M. Waller. 'Friendship Affects Gaze Following in a Tolerant Species of Macaque, *Macaca nigra*'. *Animal Behaviour* 83 (2012): 459–67. <https://doi.org/10/c4f8n2>.
- Thierry B. Where do we stand with the covariation framework in primate societies? *Am. J. Biol. Anthropol.* 128 (2021): 5–25. <https://doi.org/10.1002/ajpa.24441>
- Tomasello, M., J. Call, and B. Hare. 'Five Primate Species Follow the Visual Gaze of Conspecifics'. *Animal Behaviour* 55 (1998): 1063–69. <https://doi.org/10/bmq7xh>.
- Teufel, C., A. Gutmann, R. Pirow, and J. Fischer. 'Facial Expressions Modulate the Ontogenetic Trajectory of Gaze-Following among Monkeys'. *Developmental Science* 13 (2010): 913–22. <https://doi.org/10/b6j5r7>.